

Subclavian Artery: Anatomic Review and Imaging Evaluation of Abnormalities

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Abbreviations: MPR = multiplanar reconstruction, TOS = thoracic outlet syndrome

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Discuss the normal anatomy of the subclavian artery and its anatomic variants.
- Describe the various manifestations and causes of subclavian artery abnormalities.
- Identify expected postoperative changes associated with vascular access of the subclavian artery and distinguish them from true vascular abnormalities.

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The subclavian artery is an important structure that may be overlooked at CT of the chest and neck, in part because of its anatomic location at the periphery of the field of view but also because the clinical indication for CT examinations infrequently directs attention specifically to evaluation of the subclavian artery. As with all arteries, the subclavian artery has the potential to be involved in a variety of abnormalities, including pseudoaneurysms, dissections, stenosis or thrombosis, and vasculopathies. In addition, the subclavian artery can be secondarily involved as a collateral pathway because of an abnormality elsewhere. The subclavian artery may also be involved in surgical procedures to supply blood to other sites in the body or as an arterial access site. In these cases, recognizing the postsurgical appearance of the subclavian artery has become increasingly complex because of the use of the artery in an increasing number of procedures. Recognizing expected postoperative changes remains important to avoid mistaking them for abnormalities. The authors describe the imaging appearance of the normal anatomy of the subclavian artery and its anatomic variants, related abnormalities, and important postsurgical considerations.

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Introduction

As with all arteries, the subclavian artery can be involved in a spectrum of abnormalities. The abnormal subclavian artery can manifest with various imaging findings including stenosis, occlusion, aneurysm, pseudoaneurysm, transection, and mural inflammation. The underlying cause of these abnormalities is also variable and includes atherosclerosis, inflammatory and noninflammatory vasculopathies, and trauma. The subclavian artery can also be secondarily involved in processes centered elsewhere, particularly when it serves as a collateral pathway. Recognition of involvement of the subclavian artery in these conditions facilitates understanding of the CT examination and can aid in differential diagnosis, when necessary.

Despite its involvement in a number of important abnormalities, the subclavian artery may be overlooked or difficult to evaluate at CT. There are likely many reasons for this, such as a lower threshold for concern when the subclavian artery is not the specific reason for performing the examination or because the anatomic location of the subclavian artery is at the periphery of the field of view in CT examinations of the neck or chest. Furthermore, evaluation of the subclavian artery can be complicated by technical considerations including motion, the phase of contrast material administration, and streak artifacts from an adjacent venous contrast material column or from nearby metallic objects such as a pacemaker or shoulder arthroplasty. To avoid missing abnormalities of the subclavian artery, radiologists must evaluate it in every examination in which it is included.

The purpose of this article is to explore the imaging appearance of the subclavian artery, with an emphasis on CT. Normal anatomy,

TEACHING POINTS

- The abnormal subclavian artery can manifest with various imaging findings including stenosis, occlusion, aneurysm, pseudoaneurysm, transection, and mural inflammation. The underlying cause of these abnormalities is also variable and includes atherosclerosis, inflammatory and noninflammatory vasculopathies, and trauma.
- Despite its involvement in a number of important abnormalities, the subclavian artery may be overlooked or difficult to evaluate at CT. There are likely many reasons for this, such as a lower threshold for concern when the subclavian artery is not the specific reason for performing the examination or because the anatomic location of the subclavian artery is at the periphery of the field of view in CT examinations of the neck or chest.
- The subclavian arteries travel through the superior thoracic aperture, also known as the thoracic inlet or outlet, exit the thorax between the anterior and middle scalene muscles, and ultimately pass between the first rib and the clavicle.
- The subclavian arteries and their branches are involved in multiple collateral vascular networks in the head and neck, upper extremities, and chest. As such, they have the potential to supply collateral flow in cases of severe arterial stenosis or occlusion, whether from atherosclerotic disease, prior injury or inflammation, or extrinsic compression of the vessel.
- Postprocedural changes should be considered in the presence of adjacent surgical clips, suture material, or median sternotomy wires. Recognizing altered anatomy also helps to guide the radiologist to look for additional evidence of an arterial bypass or ligation.

anatomic variants, and important primary and secondary pathologic processes that involve the subclavian artery are reviewed. Additional attention is directed at appearances of the subclavian artery that may be affected by surgical procedures or after cannulation for bypass procedures. After reading this article, the radiologist will have an improved understanding of the subclavian artery and be well positioned to improve its evaluation at imaging.

Anatomy

Normal Anatomy

The right subclavian artery usually branches from the right brachiocephalic artery, which arises as the first branch vessel from a left aortic arch. The left subclavian artery usually arises as the third and final arch branch vessel (1). There is some variability in the configuration of the aortic arch in patients with normal branching patterns. Most notably, in patients with a bicuspid aortic valve, the left subclavian artery may have a more vertical orientation, and the distance between the origins of the left common carotid and left subclavian arteries may be increased (Fig 1).

The subclavian arteries travel through the superior thoracic aperture, also known as the thoracic

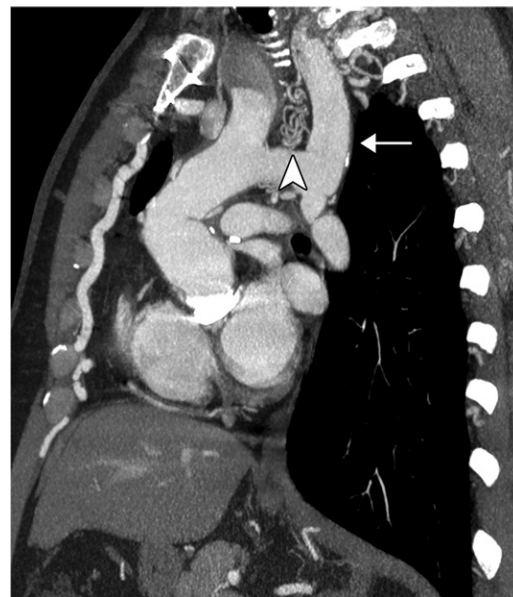


Figure 1. Left subclavian artery in a 46-year-old man with a history of bicuspid aortopathy who underwent replacement of the aortic valve and the ascending aorta. Sagittal multiplanar reconstruction (MPR) CT image shows the vertical orientation (arrow) of the left subclavian artery and the increased spacing (arrowhead) between the origins of the left common carotid artery and the left subclavian artery.

inlet or outlet, exit the thorax between the anterior and middle scalene muscles, and ultimately pass between the first rib and the clavicle (1,2) (Fig 2). The subclavian artery can be divided into three parts according to its relationship to the anterior scalene muscle. The first part of the subclavian artery extends from the origin of the subclavian artery to the medial border of the anterior scalene muscle. The second part is located posterior to the anterior scalene muscle, and the third part courses from the lateral border of the anterior scalene muscle to the lateral border of the first rib (3).

The superior thoracic aperture is demarcated posteriorly by the T1 vertebral body, laterally by the first rib, and anteriorly by the manubrium (1,2). The subclavian vessels and brachial plexus are protected by the structures in the thoracic inlet as they pass through to supply the upper extremities (4) (Figs 3, 4). Familiarity with the superior thoracic aperture is critical for accurate evaluation of the lower neck and subclavian artery. Injuries to one structure should raise concern for injury to an adjacent structure and should prompt the radiologist to examine the surrounding area closely for signs of trauma.

Anatomic Variations

The most common congenital anomaly of the aortic arch is a left-sided aortic arch with an aberrant right subclavian artery, which is present in

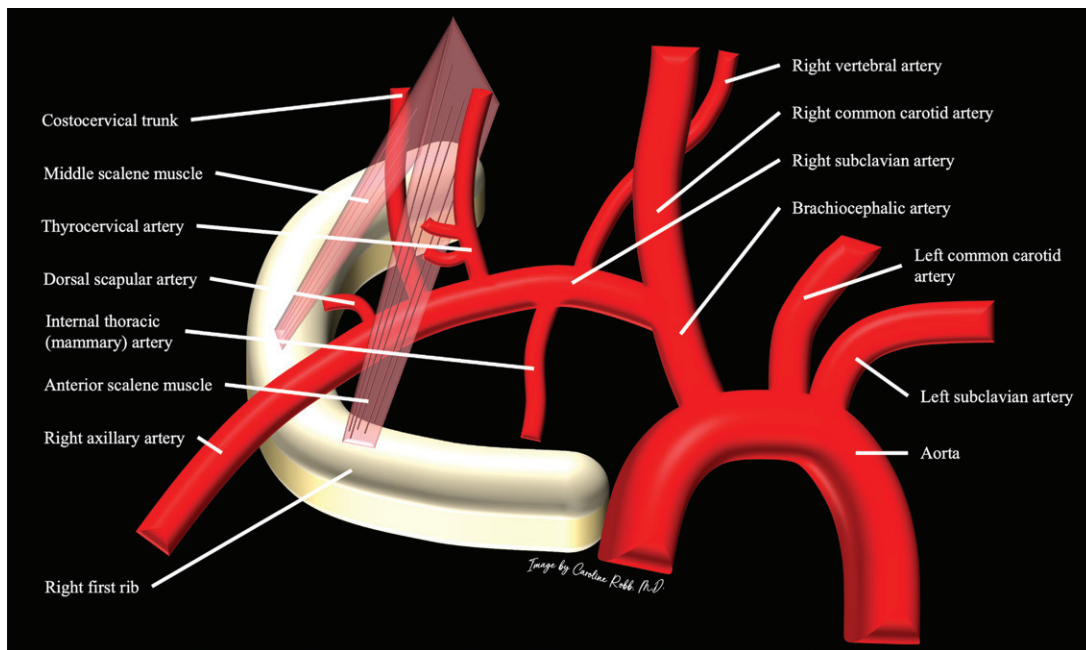


Figure 2. Illustration shows the branches of the aorta and the subclavian artery.

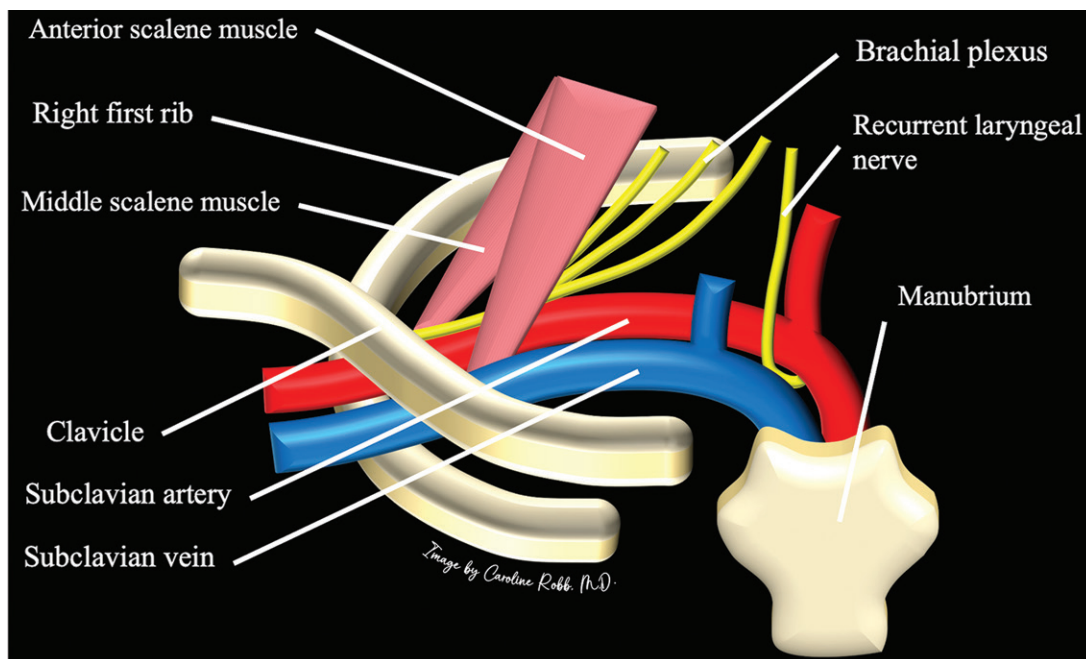


Figure 3. Illustration shows the structures in the thoracic inlet.

approximately 0.5%–2.0% of the population (5–7). The aberrant right subclavian artery arises as the last arch branch and takes a retroesophageal course. Although most patients are asymptomatic, esophageal compression causes dysphagia, which is termed *dysphagia lusoria*, in approximately 10% of patients (8) (Fig 5). In 15%–30% of cases, a retroesophageal outpouching, which is termed a *diverticulum of Kommerell*, is seen at the origin of the aberrant right subclavian artery (7) (Fig 6). A diverticulum of Kommerell is thought

to be related to the component of the primordial arch that forms the right-sided ductus arteriosus and is a strong predictor that the ligamentum arteriosum is on the contralateral side of the arch, thereby completing a loose vascular ring (4,6).

A right-sided aortic arch with an aberrant left subclavian artery is the most common variant of the right aortic arch. Similar to a left aortic arch with an aberrant right subclavian artery, the aberrant left subclavian artery is the last branch to arise from the arch and takes an oblique retroesophageal

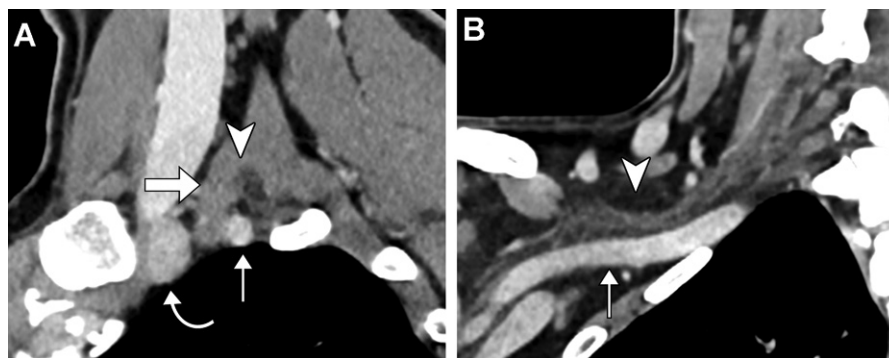


Figure 4. Contrast-enhanced sagittal (A) and coronal (B) CT images show the relationship of the structures in the thoracic inlet. Note the position of the subclavian artery (thin straight arrow) in the scalene triangle, between the anterior and middle scalene muscles, as well as the subclavian vein (curved arrow), which is located anterior to the anterior scalene muscle (thick arrow), outside the scalene triangle. The brachial plexus (arrowhead) also courses between the anterior and middle scalene muscles, abutting the posterior wall of the subclavian artery. These structures exit the superior thoracic aperture by passing between the clavicle and the first rib.

Figure 5. Aberrant right subclavian artery in a 62-year-old woman with dysphagia. (A) Esophagram shows a smooth indentation along the posterior wall of the mid-esophagus (arrow), reflecting compression of the esophagus by the retroesophageal aberrant artery, which is consistent with dysphagia lusoria. (B) Sagittal MPR CT image shows compression of the posterior wall of the mid-esophagus by the retroesophageal aberrant right subclavian artery (arrow).



course through the mediastinum to supply the left upper extremity (4,5,7). In approximately 60% of cases, there is a retroesophageal outpouching at the origin of the aberrant artery (Fig 7). Although this retroesophageal diverticulum is not technically a left-sided diverticulum of Kommerell, the term diverticulum of Kommerell is often used in this setting as well.

A right aortic arch with mirror image branching, specifically a left brachiocephalic artery arising as the first branch and right common carotid and right subclavian arteries arising as the second and third branches, respectively, is the second most common right aortic arch anomaly (4) (Fig 8). A right aortic arch with an isolated left subclavian artery is rare and is associated with congenital heart disease in approximately 50% of cases (6) (Fig 9). In these patients, the left subclavian artery connects to the pulmonary artery, either

through a patent ductus arteriosus or the ligamentum arteriosum. This variant is important to recognize because it may lead to subclavian steal syndrome and vertebrobasilar insufficiency (6).

A double aortic arch results from persistence of the left and right primordial arches, with each arch giving rise to the ipsilateral carotid and subclavian arteries (4) (Fig 10). Children may present with tachypnea, cyanosis, dysphagia, or wheezing and stridor exacerbated by crying (6,7).

Manifestations of Subclavian Artery Abnormalities

Dilatation

Aneurysm.—According to Kaufman (9), the diameter of normal subclavian arteries is 0.7–1.0 cm when they are measured orthogonal to the

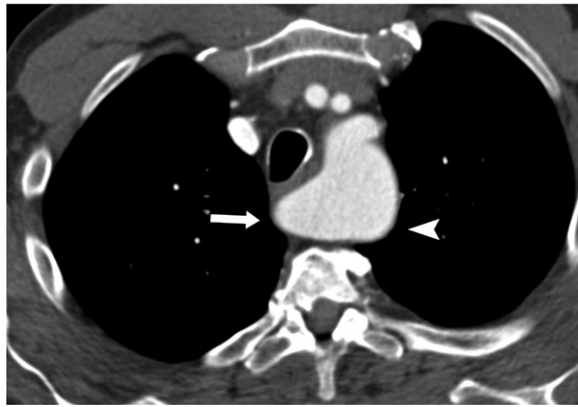


Figure 6. Aberrant right subclavian artery in a 64-year-old man. Axial MPR CT image shows the retroesophageal course of the aberrant right subclavian artery (arrow), arising as the last arch branch from a left-sided aortic arch, and an outpouching (arrowhead), arising from the origin of the aberrant artery, which is consistent with a diverticulum of Kommerell.

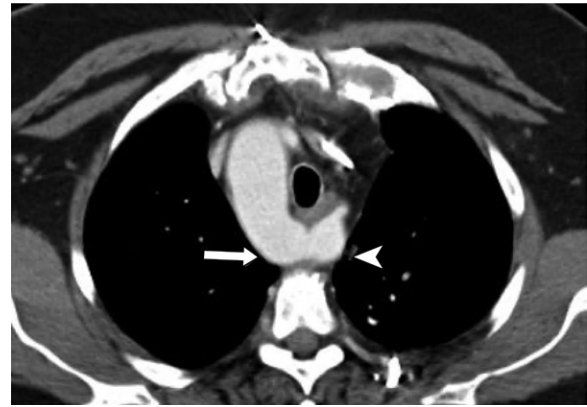


Figure 7. Aberrant left subclavian artery in a 65-year-old man. Axial MPR CT image shows the retroesophageal course of the aberrant left subclavian artery (arrow), which arises as the last branch of a right-sided aortic arch. A small retroesophageal diverticulum (arrowhead) arises near the origin of the aberrant artery.



Figure 8. Right-sided aortic arch and mirror image branching in a 50-year-old woman. Coronal MPR CT image shows a left brachiocephalic artery (arrow) that arises as the first arch vessel from the right-sided aortic arch, followed by the right common carotid and right subclavian arteries, which arise as the second and third arch vessels, respectively.

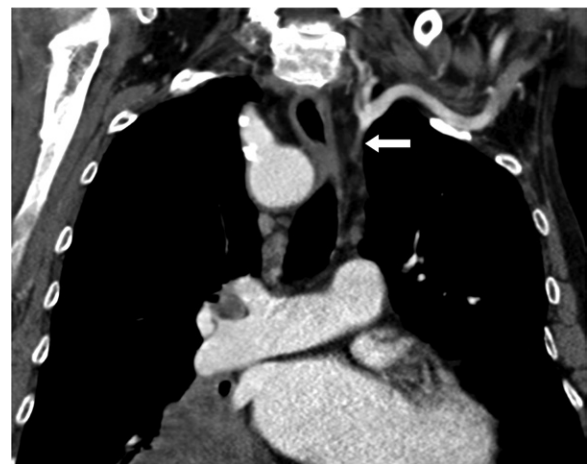


Figure 9. Isolated left subclavian artery in a 76-year-old man with a right aortic arch. Coronal CT image shows a right-sided aortic arch and an atretic cord (arrow) in the expected location of the left subclavian artery, which are findings consistent with an isolated left subclavian artery.

long axis of the vessel. With an upper threshold of 50% greater than normal size, subclavian arteries greater than 1.5 cm in diameter can be considered aneurysmal (10). True vascular aneurysms involve all layers of the vascular wall. Atherosclerosis is the most common cause of subclavian artery aneurysms. Syndromes that result in advanced and premature medial degeneration, such as Marfan syndrome and other connective tissue disorders, are other potential causes of subclavian artery aneurysms (11).

Pseudoaneurysm.—A *pseudoaneurysm* is defined as a localized dilatation or focal outpouching of a vessel wall that is contained by only some of the layers of the vessel wall, frequently only the outer-

most layer, the tunica adventitia (12,13). Trauma, including that with iatrogenic causes, and infection are important causes of subclavian pseudoaneurysms (12,14). Imaging findings seen in patients with subclavian artery pseudoaneurysms include eccentric outpouching, an irregular contour, partial or complete thrombosis, and, in cases of traumatic injury, hemorrhage adjacent to and in contact with the damaged artery. In cases of rupture, intravenous contrast extravasation into adjacent tissues may be seen (13). Infected subclavian artery pseudoaneurysms may be associated with lymphadenopathy and inflammation of the adjacent fat.

Stenosis and Occlusion

Thrombosis of the subclavian artery most commonly occurs in patients with progressive

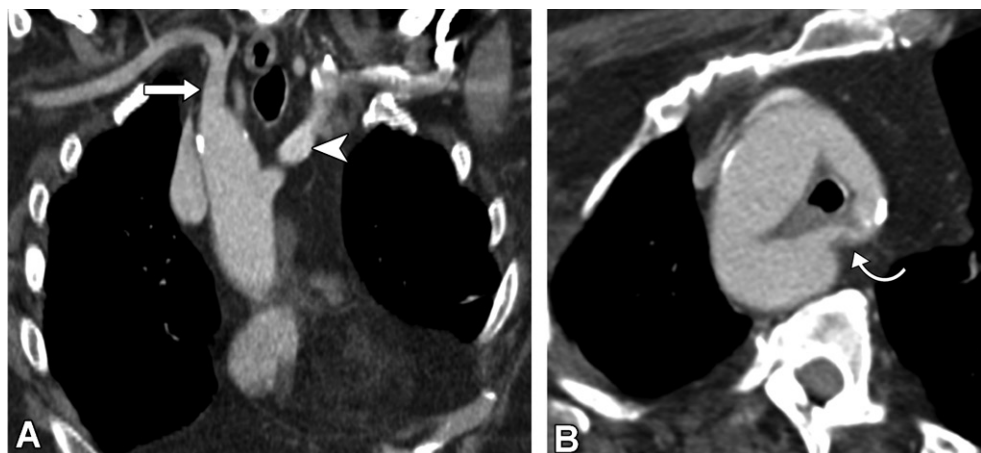


Figure 10. Double aortic arch in a 73-year-old woman. (A) Coronal MPR CT image shows a right brachiocephalic artery (arrow) arising from the right arch and the left subclavian artery arising from the left arch (arrowhead). (B) Oblique axial CT image shows the narrow vascular bridge (curved arrow) posterior to the trachea that connects the two arches and completes a vascular ring.

atherosclerosis, but it can also occur as a result of injury, hypercoagulable states, or embolism (5,15). Progressive chronic thrombosis is more common than acute thrombosis, allowing patients to develop collateral vasculature in many cases (5). In cases of severe stenosis or occlusion, particularly when acute, patients may present with symptoms of claudication, including acute-onset hand or upper extremity pain (5,15). Thrombosis of the subclavian artery can also place the patient at risk for distal emboli to the ipsilateral upper extremity. Management options include anticoagulation therapy, thrombolysis or thrombectomy, placement of an endovascular stent, and surgical bypass.

Dissection

Dissections are characterized by an intimal defect that allows blood to dissect along the media, creating an intimal flap that separates the true and false lumina (11). The intimal flap is usually identifiable in subclavian artery dissections (Fig 11). However, because of the small caliber of the vessel, the false lumen may collapse or thrombose, giving the appearance of an eccentric smooth occlusion rather than an intimal flap. This appearance, particularly when it is associated with an aortic dissection or in patients with atherosclerosis, should raise concern for a subclavian artery dissection. Subclavian artery dissection is most commonly seen in patients with an aortic dissection that extends into one or both subclavian arteries (5). Isolated subclavian artery dissection is rare; most cases are associated with arterial thoracic outlet syndrome (TOS), connective tissue disease, blunt trauma, or iatrogenic injury. Patients may present with chest or back pain, arm paresthesia, or nonspecific neurologic symptoms. Dissection predisposes patients to formation of a thrombus, which may result in a distal embolism.

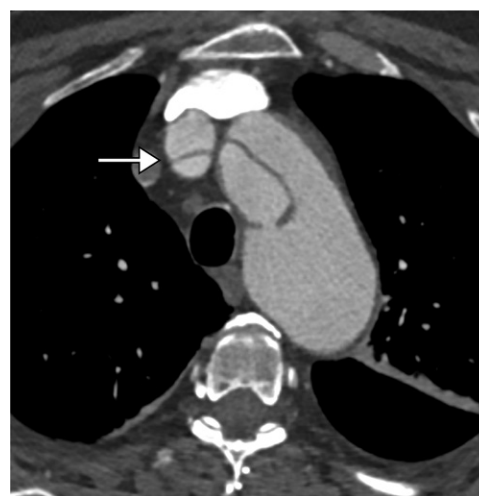


Figure 11. Dissection flap in a 79-year-old man who presented with acute chest pain and hypertension. Axial CT image shows a dissection flap involving the ascending aorta and right brachiocephalic artery (arrow).

Causes of Subclavian Artery Abnormalities

Thoracic Outlet Syndrome

TOS refers to the symptoms associated with compression of the neurovascular bundle that supplies the upper extremity and may have an arterial, venous, or neurogenic cause (16). Most patients with TOS participate in an activity that involves an overhead arm motion, which increases the frequency and degree of compression of the structures of the thoracic outlet (2).

Arterial TOS, which is caused by compression and subsequent damage of the subclavian artery, is the least common of the three subtypes of TOS. The symptoms are related to distal ischemia of the upper extremity, which is often the

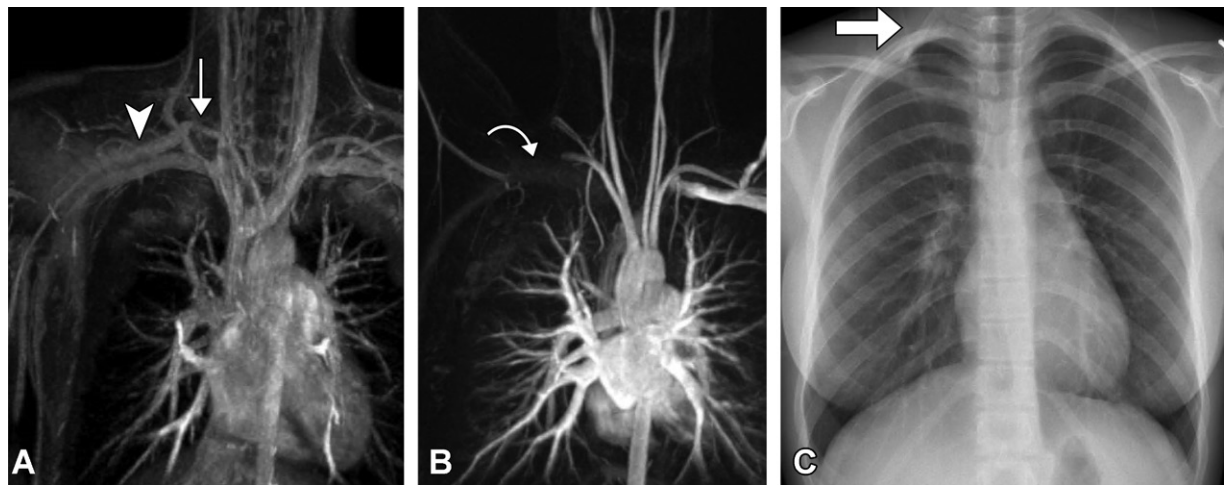


Figure 12. Arterial thoracic outlet syndrome in a 14-year-old adolescent girl with a painful right neck nodule, a bruit, and right arm pain that worsened with exercise. **(A)** Maximum intensity projection MR angiogram acquired with the right arm at the patient's side shows focal narrowing (arrow) of the right subclavian artery with poststenotic dilatation (arrowhead). **(B)** Maximum intensity projection MR angiogram acquired with the right arm raised shows occlusion of the artery at the site of stenosis (curved arrow). **(C)** Frontal chest radiograph shows a right cervical rib (arrow).

Imaging Findings of Arterial TOS

Axillo-subclavian arterial aneurysm or pseudoaneurysm
Axillo-subclavian arterial dynamic narrowing or stenosis
Arterial thrombus
Distal emboli
Enlarged collateral vessels
Bone or soft-tissue abnormalities
Cervical rib
Elongated C7 transverse process
Anomalous first rib
Old fracture

result of distal emboli originating from a damaged subclavian artery, and may include pain, weakness, and a cold limb (5). Arterial TOS is associated with either a rib anomaly (cervical or anomalous rib) or rib fracture in most cases (2). As such, it may be best considered to be the product of an anatomic predisposition and an exacerbating factor (16).

Imaging includes MRI or CT angiography performed with the patient's arm positioned by their side and in a raised position. CT and MRI show subclavian artery narrowing that worsens with arm abduction in many patients with arterial TOS (2,16) (Fig 12). However, this finding is not exclusively sufficient for diagnosis, as many patients without clinical TOS have similar imaging findings. More important imaging findings include those of vascular damage, specifically, a subclavian artery aneurysm or pseudoaneurysm, thrombosis that is not due to atherosclerosis at the site of compression, a fixed stenosis, or an embolism (16). Imaging findings of arterial TOS are summarized in the Table.

Subclavian Steal Syndrome

The subclavian artery may “steal” blood supply by means of reversal of flow in a branch vessel when there is high-grade stenosis in the subclavian artery proximal to the origin of that branch vessel. *Subclavian steal syndrome* is defined by the associated clinical manifestations, which may include limb ischemia, asymmetric blood pressure in the upper extremities, and neurologic symptoms (4,5).

The most commonly involved vessel is the vertebral artery. When the vertebral artery is involved, vertebrobasilar insufficiency may result from reversal of blood flow from the intracranial circulation to reconstitute the occluded subclavian artery (5). The diagnosis can be suggested when a focal upstream stenosis is seen in the subclavian artery in a patient with appropriate symptoms. Reversal of vertebral artery flow at Doppler US or phase-contrast MRI allows confirmation of the diagnosis (15) (Fig 13).

Coronary subclavian steal syndrome is a rare variation that may be seen when the left internal mammary artery has been used in a coronary artery bypass graft procedure (5,17). In these cases, the subclavian artery receives retrograde blood flow through the left internal mammary artery (17) (Fig 14). Patients may present with chest pain and signs of myocardial ischemia (5). In patients with left internal mammary bypass grafts, radiologists should carefully interrogate the graft and the proximal subclavian artery because stenosis in either entity can limit flow to the territory of the bypassed coronary artery.

Injury

Traumatic Injury.—Isolated acute traumatic subclavian artery injury is rare, accounting for

Figure 13. Subclavian steal syndrome in an 85-year-old woman who presented with unequal radial pulse and altered mental status. (A) Coronal CT image shows mixed calcified and noncalcified atherosclerotic plaque of the proximal left subclavian artery that results in severe stenosis (arrow). (B) Time-of-flight MR angiogram of the head shows a lack of antegrade flow in the left vertebral artery (arrowhead). (C) Doppler US image shows retrograde blood flow through the left vertebral artery.

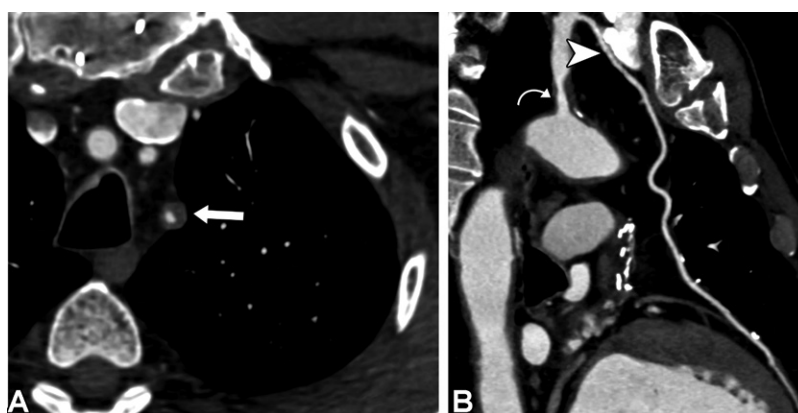
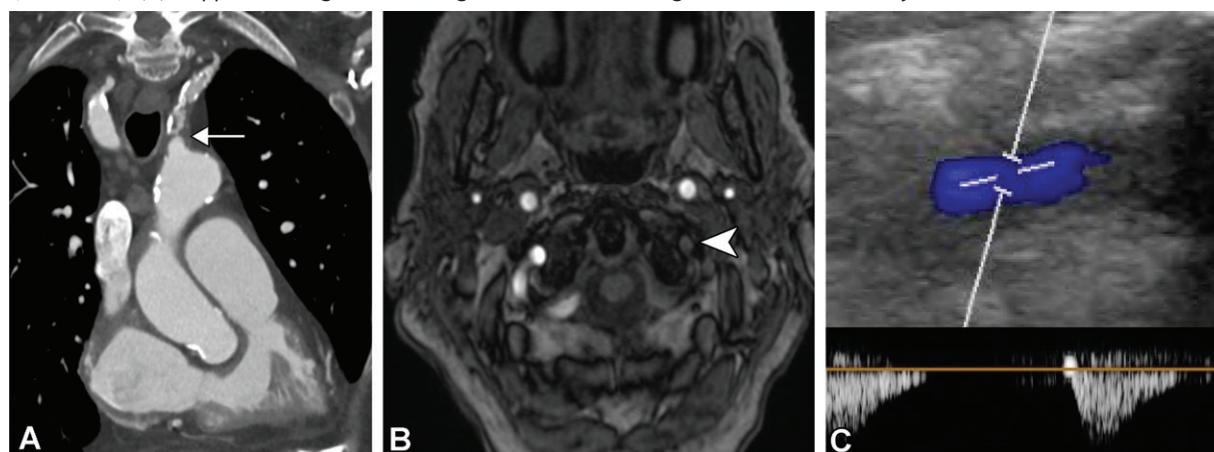


Figure 14. Subclavian steal in a 63-year-old man with a history of a coronary artery bypass graft procedure, which included bypass of the left internal mammary artery to the left anterior descending artery. The patient presented with chest pain, an elevated high-sensitivity troponin level (>93 ng/mL [>93 $\mu\text{g/L}$]), and imaging findings that suggest myocardial ischemia related to subclavian steal, involving the left internal mammary artery. (A) Axial CT image shows severe stenosis of the proximal left subclavian artery (arrow). (B) Curved MPR CT image also shows severe stenosis of the proximal left subclavian artery (curved arrow) before the origin of the left internal mammary artery, which remains patent throughout its course (arrowhead).

only 1%–2% of reported acute vascular injuries. The low incidence is primarily due to the location of the subclavian artery in the chest, where it is protected by surrounding osseous and muscular structures (3). When subclavian artery injury is suspected, noncontrast and/or delayed portal venous phase CT images should be obtained in addition to arterial phase CT images to better distinguish nonvascular sources of hyperattenuation and better characterize vascular injuries and extravasation, if they are present.

Penetrating Injury.—Most subclavian artery injuries are due to penetrating trauma, most often from gunshot wounds and knife injuries (18). Identification of the wound trajectory is essential; any

trajectory in the vicinity of the subclavian artery should increase suspicion for injury. Direct findings of injury include abrupt occlusion, extravasation of contrast material, outpouching of the vessel, or formation of a pseudoaneurysm. Of these, contrast extravasation is the least commonly observed (Fig 15). In some cases, localized tissue disruption can cause vasospasm and local occlusion that may simulate an occlusion due to vascular injury.

Blunt Injury.—Blunt traumatic injury to the subclavian artery is estimated to account for only 1%–3% of acute subclavian artery injuries, which is likely an underestimate, given the highly fatal nature of the traumatic events that cause these types of injuries (3). A high-force impact that is strong

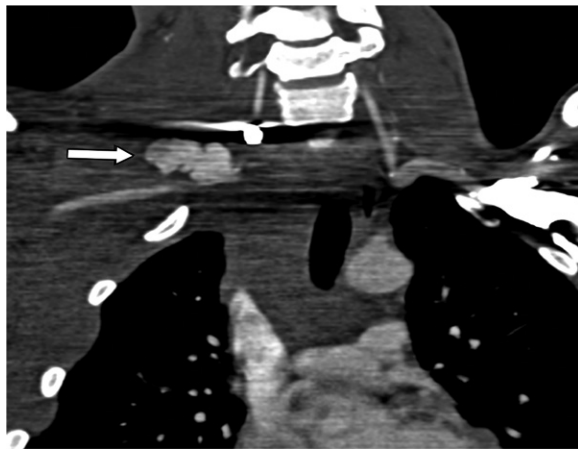


Figure 15. Right subclavian artery injury in a 22-year-old man who presented with a gunshot wound to the right shoulder. Coronal CT image shows a retained bullet in the soft tissues of the right side of the neck and an irregular outpouching of contrast material (arrow) arising from the right subclavian artery, which is consistent with an acute traumatic injury, with a pseudoaneurysm and extravasation.

enough to cause rapid deceleration or fractures of the surrounding bones is required to injure the subclavian artery in the absence of penetrating trauma. Most patients with blunt traumatic injury to the subclavian artery present with polytrauma from a motor vehicle crash, including automobiles, motorcycles, and all-terrain vehicles (18).

CT findings of subclavian artery injury are the same as those for any other major vessel. Hematoma in direct contact with the subclavian artery should be considered an indirect sign of injury. Direct signs of injury include focal outpouching or a pseudoaneurysm, an abrupt caliber change, thrombosis, and, in severe cases, contrast material extravasation (Fig 16). Radiologists must take care to avoid “satisfaction of search” in patients with trauma and carefully evaluate the subclavian arteries for injury. Isolated subclavian artery injuries, as with those of all the great vessels, tend to manifest with mediastinal hematomas that are more cranial than those associated with traumatic aortic injury. The presence of a “high” mediastinal hematoma may be a clue to look for these challenging injuries. Dedicated multiplanar reconstruction (MPR) CT images are helpful in characterizing and describing the extent of subclavian artery injuries. Additional injuries may include rib fracture, clavicle fracture, pneumothorax, pulmonary contusion, traumatic brain injury, or damage to the brachial plexus (18). Regardless of the mechanism of injury, prompt diagnosis and intervention are essential, as subclavian artery injury may be associated with major complications including severe hemorrhage, hemothorax, formation of a pseudoaneurysm, cerebrovascular infarct, permanent occlusion, or exsanguination (18).

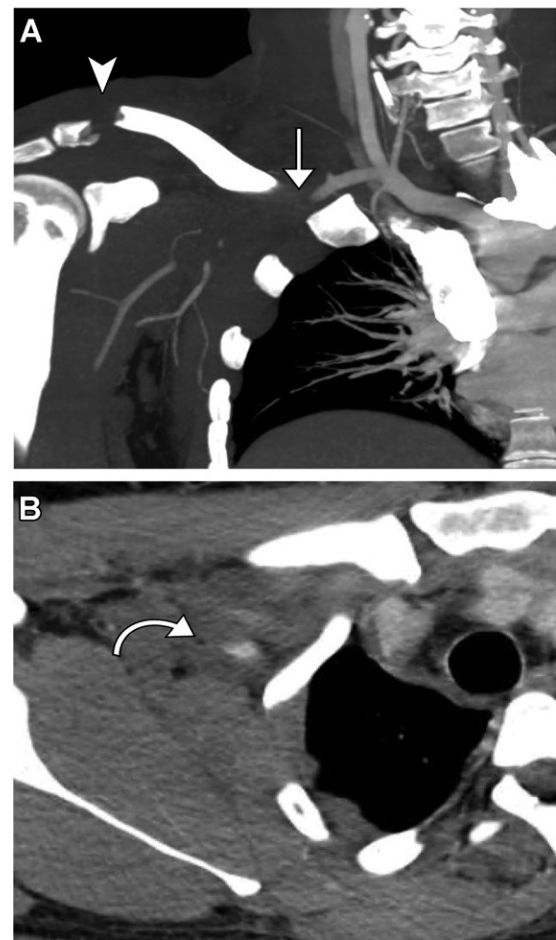


Figure 16. Right subclavian artery injury in a 37-year-old man who presented with paralysis, decreased sensation, and pulselessness of the right upper extremity after an all-terrain vehicle accident. (A) Coronal MPR CT angiogram shows an abrupt cutoff of the right subclavian artery and the axillary artery (arrow), with downstream opacification of the distal axillary artery via collateral vessels from the brachial artery. The right distal clavicle is fractured (arrow-head). (B) Axial CT angiogram shows extensive soft-tissue stranding and a hematoma surrounding the right subclavian artery that extends along the right axillary artery and branches (curved arrow).

Iatrogenic Injury.—The subclavian artery may also be involved in iatrogenic injury. Inadvertent arterial cannulation of the subclavian artery is rare and occurs in less than 1% of central venous catheter placements (19). However, when this does occur, the radiologist plays a vital role in the patient’s care.

Knowledge of the anatomic position of the subclavian artery with respect to the subclavian vein and the superior vena cava, particularly on chest radiographs, helps the radiologist to identify these mistakes quickly and accurately. The scalene tubercle is a small projection along the medial margin of the first rib, at the insertion of the anterior scalene muscle on the first rib. Its approximate position can be seen on a frontal view image of the chest, where the first rib overlaps with the

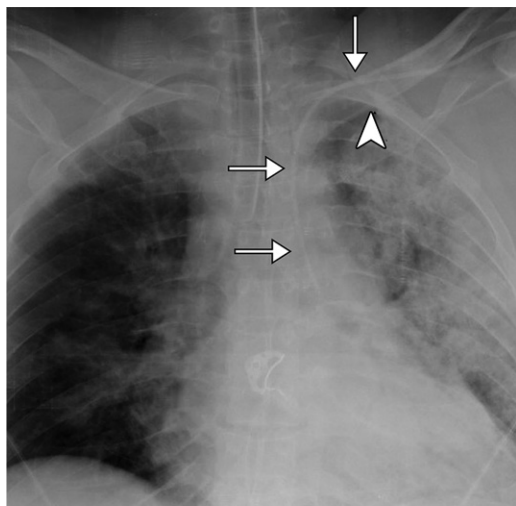


Figure 17. Inadvertent placement of a central venous catheter into the left subclavian artery in a 53-year-old man. Anteroposterior chest radiograph shows the catheter (vertical arrow) coursing cranial to the scalene tubercle (arrowhead), in the region where the first rib and clavicle overlap. Note the central course of the catheter (horizontal arrows) as it follows the descending thoracic aorta.

clavicle. The subclavian vein travels anterior and slightly inferior to the anterior scalene muscle, while the subclavian artery travels posterior and slightly superior to it (3) (Fig 17). Thus, on chest radiographs, catheters that course cranial to the expected location of the scalene tubercle should prompt a query for incorrect placement in the subclavian artery rather than in the subclavian vein. An additional clue to arterial placement involves the catheter following the course of the aorta more centrally. Prompt identification of potential arterial catheters and urgent consultation with a vascular surgeon is critical to reducing the chances of substantial harm to the patient. These lines should not be manipulated or removed at the bedside because such actions may result in substantial morbidity. Arterial placement can be confirmed by measurement of vascular pressure or at CT. CT also allows confirmation of the point of entry into the artery, which will help to guide management.

Vasculopathy

Infectious Vasculitis

Infectious vasculitis is caused by direct damage of the vascular wall by an infectious pathogen and often results in the development of an infectious pseudoaneurysm. At imaging, infectious pseudoaneurysms are often saccular and may have a narrow neck, which is related to their previous description as “mycotic” aneurysms. (13) (Fig 18). Because they develop rapidly, infectious

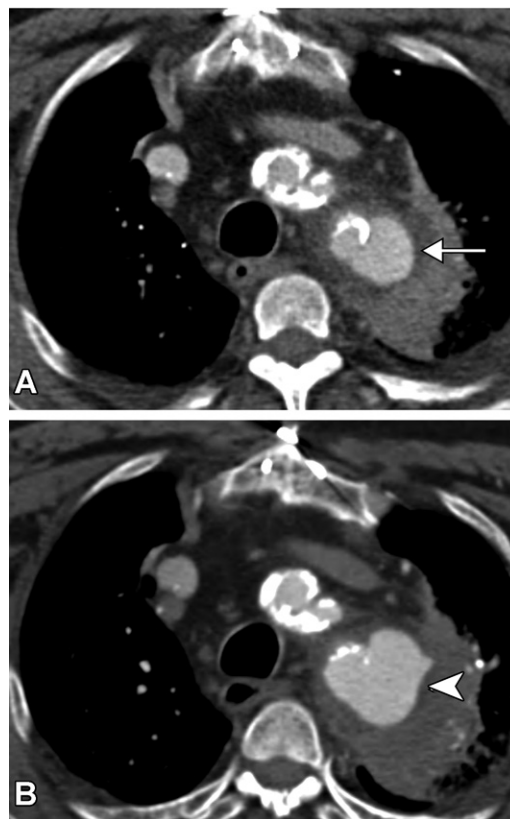


Figure 18. Infectious pseudoaneurysm of the origin of the left subclavian artery in a 56-year-old man. (A) Axial CT image shows a saccular outpouching (arrow) arising from the proximal left subclavian artery. Note the narrow neck, adjacent fat stranding, and the lack of peripheral calcification. (B) Axial CT image acquired 4 days later shows rapid enlargement of the outpouching (arrowhead), which is consistent with an infectious pseudoaneurysm.

pseudoaneurysms are not calcified. There may be associated lymphadenopathy and inflammation of the adjacent surrounding fat. Infectious pseudoaneurysms are unstable and can rapidly enlarge over the course of days.

The imaging features of infectious pseudoaneurysms overlap with those of unstable sterile aneurysms and pseudoaneurysms, thereby necessitating the correlation of suspicious imaging findings with the patient's symptoms at clinical presentation. Patients typically present with bacteremia, fever, and other constitutional signs and symptoms of infection (20). The suggestion of a possible infectious pseudoaneurysm has a substantial effect on patient treatment, including the prompt and prolonged use of antibiotics and consideration of open surgical débridement and resection rather than placement of an endovascular device, which would then be at risk for infection.

Inflammatory Vasculitis

Inflammatory vasculitides are diseases that result in inflammation of vessel walls in the absence

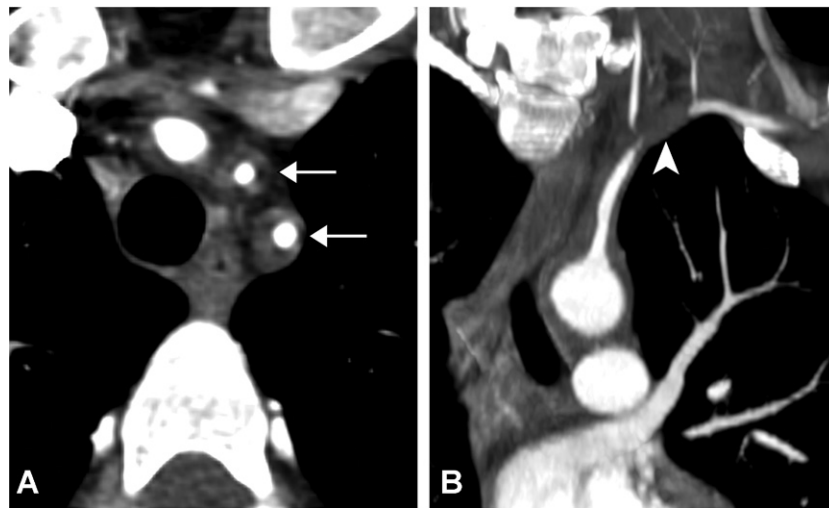


Figure 19. Occlusion of the left subclavian artery in a 29-year-old woman who presented after 4 months of intermittent left arm numbness, weakness, and color changes and eventually received a diagnosis of Takayasu arteritis. Axial (A) and coronal (B) CT angiograms show circumferential long-segment mural thickening around the aortic arch that extends along the great vessels (arrows in A), resulting in multifocal tapered stenoses and occlusion of the left subclavian artery (arrowhead in B).

of an infectious agent. According to the 2012 International Chapel Hill Consensus Conference on the Nomenclature of Systemic Vasculitides (21), vasculitides are subdivided according to the size of vessel that is most likely to be affected, and the subclavian artery is considered a large vessel. Although any inflammatory vasculitis can involve a vessel of any size, the two most common idiopathic large vessel vasculitides are Takayasu arteritis and giant cell arteritis. Both diseases are characterized by idiopathic granulomatous transmural inflammation and arterial wall thickening of the aorta and arch branches, including the subclavian artery, that often leads to arterial thrombosis, stenosis, and aneurysm (5). These findings can be detected at CT and MRI. Fluorine-18 fluorodeoxyglucose PET/CT can also be used to evaluate inflammatory activity in vasculitis before and after treatment, because vessel walls affected by active large vessel inflammatory vasculitis are metabolically active (22).

Both Takayasu arteritis and giant cell arteritis have a strong female predilection. However, Takayasu arteritis is typically seen in a younger patient population (approximately 10–40 years of age) and Asian patients, whereas giant cell arteritis is more often seen in an older patient population (approximately 70–80 years of age) and in White patients (5). Takayasu arteritis has a propensity to involve the subclavian arteries and may result in a nonpalpable brachial pulse or asymmetric blood pressure in the upper extremities, hence the moniker “pulseless arteritis” (5,23) (Fig 19). Giant cell arteritis, on the other hand, is known as temporal arteritis, given the frequent involvement of the temporal artery. Patients with giant cell arteritis may present with headache, jaw pain, unexplained fever, or visual disturbances (5). In many cases, distinguishing between Takayasu arteritis and giant cell arteritis is not possible at imaging or on the

basis of clinical grounds. In these cases, patients younger than 50 years with large vessel vasculitis are considered to have Takayasu arteritis, while those older than 50 years are described as having giant cell arteritis.

In addition to the idiopathic large vessel vasculitides, secondary large vessel vasculitides such as those due to chemotherapy or a known systemic disease, such as sarcoidosis or immunoglobulin G4-related disease, can produce identical findings and also involve the subclavian artery (Fig 20).

Fibromuscular Dysplasia

Fibromuscular dysplasia is an idiopathic nonatherosclerotic and noninflammatory condition that predominantly affects small- and medium-sized arteries, most often the renal, extracranial vertebral, and carotid arteries (24,25). The subclavian arteries can also be involved, either in isolation or in combination with involvement of other vessels. The classic finding of fibromuscular dysplasia involves short focal stenoses with alternating areas of dilatation that produces a “string of beads” appearance (24) (Fig 21). However, in some cases, an isolated focal stenosis may be the only finding. Diagnosis is reliant on the presence of at least one site of stenosis without another cause. Arterial tortuosity, dissection, and aneurysm may also be seen (24).

Connective Tissue Vasculopathies

The subclavian arteries are often involved in connective tissue vasculopathies, either in isolation or in association with findings in other vessels. A connective tissue vasculopathy should be suspected when aneurysms or extensive tortuosity of the subclavian arteries is seen in the absence of atherosclerosis, particularly in younger patients. The major syndromic connective tissue vasculopathies are briefly reviewed in the next sections.

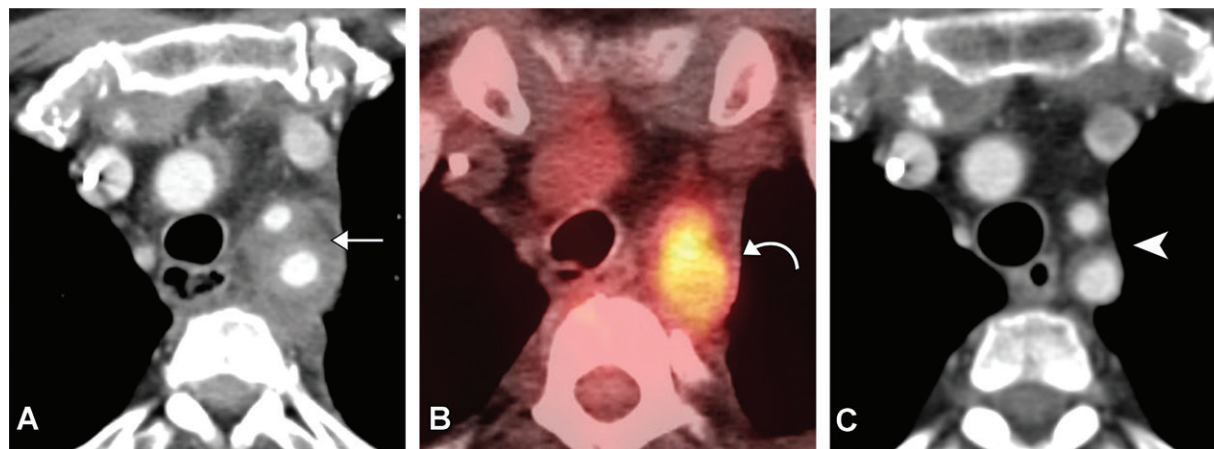


Figure 20. Cisplatin-induced vasculitis in a 64-year-old woman with a history of urinary bladder cancer who presented with a fever 2 weeks after the first cycle of cisplatin and doxorubicin. Results of the workup to evaluate for infection were negative, and temporal artery biopsy results showed no evidence of inflammation. (A) Axial CT image shows soft-tissue thickening involving the walls of the subclavian and carotid arteries (arrow). (B) Axial PET/CT image shows areas of increased fluorine-18 fluorodeoxyglucose avidity in the same areas of soft-tissue thickening, which are findings consistent with ongoing inflammation (curved arrow). (C) Axial CT image acquired 4 months after discontinuation of cisplatin shows nearly complete resolution of the arterial wall thickening and soft-tissue inflammation (arrowhead). These findings are consistent with cisplatin-induced vasculitis.

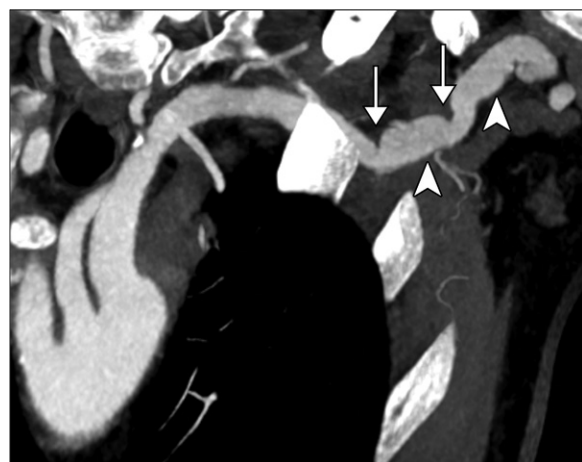


Figure 21. Fibromuscular dysplasia in a 75-year-old man who presented with back pain and received a diagnosis of type B aortic dissection. Coronal MPR CT image shows alternating areas of stenoses (arrows) and aneurysms (arrowheads) in the left subclavian artery, creating the “string of beads” appearance. Webs were seen in the vessels during surgical repair, which allowed confirmation of fibromuscular dysplasia.



Figure 22. Marfan syndrome in a 33-year-old woman. Coronal CT image shows an isolated dissection flap involving the left subclavian artery (arrow).

Of note, there are other inherited nonsyndromic vasculopathies, some of which have known genes that may also predispose patients to abnormalities of the subclavian arteries.

Marfan Syndrome.—Marfan syndrome is characterized by impaired formation of elastin, which predisposes a patient to loss of elasticity, cystic medial necrosis, fibrosis, and loss of smooth muscle in artery walls. These underlying histologic abnormalities manifest as the classic features of Marfan syndrome, specifically aortic root aneurysms involving the sinuses of Valsalva and a predilection for type A aortic dissections (5). The

subclavian arteries may be involved in isolation, typically manifesting as an aneurysm or, less commonly, as an isolated dissection. More commonly, the subclavian arteries are secondarily involved by direct extension from the aorta (11) (Fig 22).

Loeys-Dietz Syndrome.—Loeys-Dietz syndrome is a rare disease that was initially described in 2005. Five types have been described and are characterized by craniofacial and vascular abnormalities (5,26). The vascular abnormalities include aneurysms and arterial tortuosity, the latter of which is a defining feature of Loeys-Dietz syndrome (5) (Fig 23). Many features of Loeys-Dietz syndrome



Figure 23. Loeys-Dietz syndrome in a 33-year-old man. Coronal MPR CT image shows the characteristic arterial tortuosity involving the aortic arch branch vessels and a right brachiocephalic artery aneurysm (arrow).

overlap with those of Marfan syndrome, but certain imaging characteristics can help a radiologist distinguish between the two syndromes. Aortic root aneurysms do occur, but aneurysms involving arteries other than the aorta, particularly the subclavian arteries, are more common in patients with Loeys-Dietz syndrome than in those with Marfan syndrome. In Loeys-Dietz syndrome, these aneurysms occur more frequently and are more aggressive, with a tendency to develop dissections at a smaller size and in patients at a younger age than those with Marfan syndrome (5).

Ehlers-Danlos Syndrome.—Vascular Ehlers-Danlos syndrome (type IV) is rare, representing approximately 5%–10% of all patients with Ehlers-Danlos syndrome (27). These patients are at increased risk for developing aneurysms and arterial dissections because of a deficiency in type III procollagen that results in excessive tissue fragility (5,27). Medium and large vessels are most often affected, particularly the carotid and vertebral arteries; the subclavian arteries may also be involved (28) (Fig 24).

Role of Subclavian Arteries as Collateral Vessels

Thoracic Collateral Vessel

The subclavian arteries and their branches are involved in multiple collateral vascular networks in the head and neck, upper extremities, and chest. As such, they have the potential to supply collateral flow in cases of severe arterial stenosis or occlusion, whether from atherosclerotic disease, prior injury or inflammation, or extrinsic compression of the vessel. This is of particular importance in patients with cystic fibrosis, because the subclavian

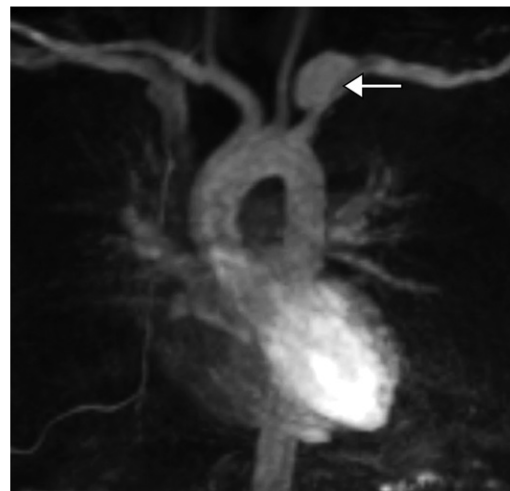


Figure 24. Vascular Ehlers-Danlos syndrome in a 39-year-old man who presented with left shoulder and back pain despite receiving optimized medical treatment. Chest MR angiogram shows an aneurysm of the left subclavian artery, which is associated with an isolated dissection flap (arrow).

artery and its branches often provide nonbronchial systemic arterial and bronchial collateral arterial flow, especially in cases of upper lobe–predominant disease (Fig 25). Awareness of these potential collateral vessels is critical for evaluating and treating cases of hemoptysis (29,30). The subclavian artery may also serve as a source of collateral vascular flow in patients with cyanotic congenital heart disease, most notably in those with repaired single ventricle morphology or with unrepaired conotruncal abnormalities. These collateral vessels often anastomose with pulmonary artery branches.

Aortic Collateral Vessel

Thoracic endovascular aortic repair is often performed for the management of type B thoracic aortic dissections and aneurysms. Depending on the location and extent of disease, the origin of the left subclavian artery may need to be covered by the stent, with the proximal landing zone of the graft located proximal to the origin of the left subclavian artery (31).

Many surgeons proactively create a subclavian-to-carotid bypass with or without ligation or occlusion of the native subclavian artery in efforts to prevent subclavian steal, vertebrobasilar insufficiency, upper extremity ischemia, and spinal cord ischemia (31,32). When the proximal native subclavian artery is not ligated, the prevention of an endoleak relies on the formation of a thrombus at the origin of the vessel (33). However, in some cases, the artery remains patent and provides retrograde filling of the excluded aorta, resulting in a type II endoleak (Fig 26). If the endoleak continues to enlarge or causes symptoms, further intervention is required.

Figure 25. Cystic fibrosis in a 46-year-old man who presented with hemoptysis and bronchial collateral circulation from the right subclavian artery. (A) Coronal MPR CT image shows multiple collateral vessels (arrow) coursing from the right subclavian artery (arrowhead), through the bronchiectatic right upper lobe, and to the right pulmonary artery. (B) Fluoroscopic angiogram with contrast material injection into the right brachiocephalic artery during bronchial artery embolization shows contrast material flowing from the right subclavian artery to the right pulmonary artery through the same collateral vessels (arrow).

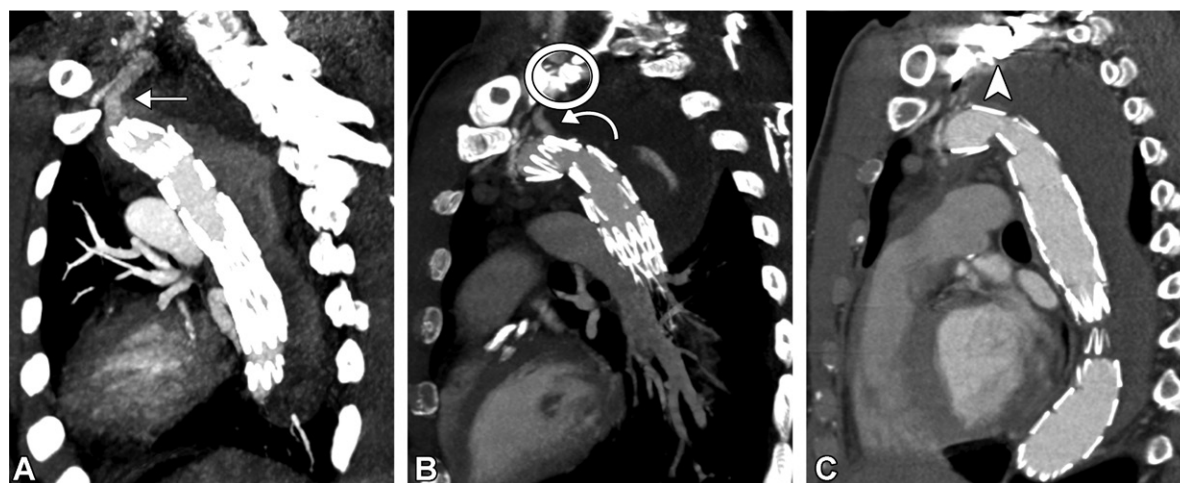
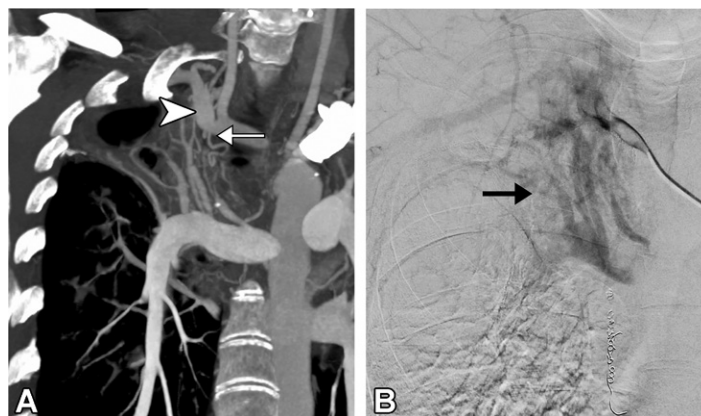


Figure 26. Type II endoleak after thoracic endovascular aortic repair in a 53-year-old man with a history of chronic type B thoracoabdominal aortic dissection and progressive aneurysmal dilatation. (A) Sagittal MPR CT angiogram shows changes after thoracic endovascular aortic repair, with the proximal landing zone of the graft located proximal to the origin of the left subclavian artery. Despite this positioning, contrast material extends from the stump of the native left subclavian artery (arrow) and opacifies a large type II endoleak along the endovascular graft. (B) Sagittal MPR CT image acquired at a later date shows attempted treatment with the placement of an Amplatzer (Abbott) vascular plug (circle). However, contrast material still opacifies the proximal native left subclavian artery by means of retrograde perfusion (curved arrow), which is consistent with a persistent type II endoleak and was later confirmed at angiography. (C) Sagittal MPR CT image shows resolution of the type II endoleak after coil embolization (arrowhead).

Arteriovenous Malformation

Arteriovenous malformations involving the subclavian artery are rare and often involve the vertebral and common carotid arteries in conjunction with the subclavian and internal jugular veins (34,35) (Fig 27). Although these arteriovenous malformations are typically asymptomatic, they should be treated to prevent complications related to high-output heart failure or spontaneous hemorrhage. Treatment is usually not curative, given the complex vascular supply (34). Surgery poses a high risk of bleeding and injury to the brachial plexus. Endovascular intervention such as embolization is less invasive and has low morbidity; however, not all arteriovenous malformations are amenable to embolization, and care must be taken to avoid embolizing arteries that supply blood to other areas of the body (35).

Mimics of Subclavian Abnormalities

Subclavian Grafts

The subclavian artery is a versatile vessel that may be used to supply alternative arterial blood flow to the head and neck, upper extremities, and lower extremities. Complex repairs of the aortic arch often require establishing an alternative route of blood supply to the upper extremities. Carotid to subclavian artery bypasses are commonly performed when a thoracic endovascular aortic repair stent covers the origin of the left subclavian artery (32) (Fig 28). Blood flow to the left upper extremity is reestablished by bypassing the origin of the subclavian artery and using a graft to connect the subclavian artery to the ipsilateral carotid artery (36). In these cases, the proximal end of the native subclavian artery may also be ligated or oversewn to prevent an endoleak.

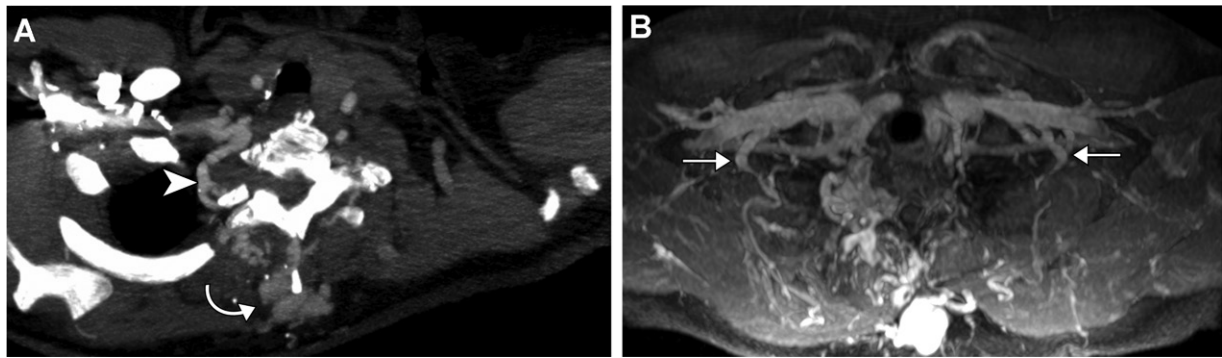


Figure 27. Arteriovenous malformation in a 53-year-old woman with a slowly growing painful lump on the upper back. (A) Oblique axial MPR CT image shows a complex high-flow arteriovenous malformation (curved arrow) involving the paraspinal muscles and the subcutaneous tissues of the right upper back. One of the large tortuous feeding arteries (arrowhead) arises from the right subclavian artery. (B) Axial MR angiogram shows the same findings as those on the CT image and better highlights the draining veins (arrows) that empty into the left and right subclavian veins.

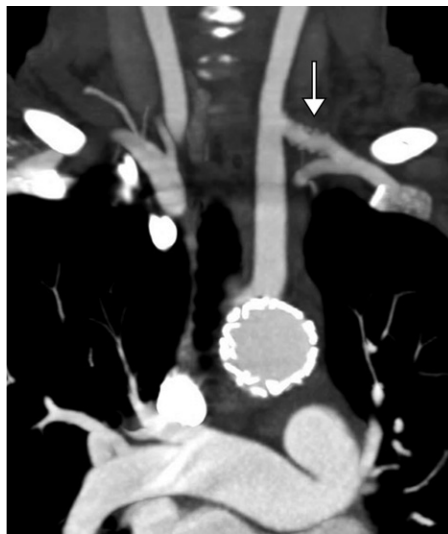


Figure 28. Changes of thoracic endovascular repair and left carotid to subclavian artery bypass in a 45-year-old woman with a history of thoracic aortic aneurysm. Coronal CT image shows a bypass graft (arrow) from the left carotid artery to the ipsilateral subclavian artery that was placed to preserve blood flow to the left upper extremity after the origin of the left subclavian artery was covered by a stent.

In cases of aortic arch repair, the presence of stenosis or dissection involving the brachiocephalic or carotid arteries may necessitate performance of an aorto-subclavian bypass (Fig 29). The subclavian artery may also be used in subclavian-femoral bypass grafts to provide blood flow to the lower extremities in cases of aortoiliac occlusive disease. In patients with end-stage renal disease for whom other options for hemodialysis access have been exhausted, the subclavian artery may be an alternative site for an arteriovenous graft (Fig 30).

Arterial Access

Side grafts, which are sometimes referred to as arterial conduits, are commonly used to secure

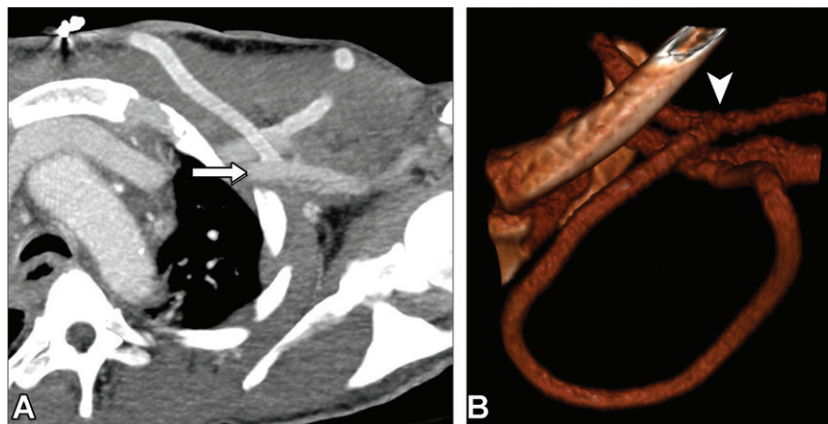


Figure 29. Changes of an ascending aortic replacement and aorto-subclavian bypass in a 44-year-old man with a history of type A aortic dissection and aortic root aneurysm. Sagittal MPR CT image shows the bypass graft (arrow) arising from the anterior wall of the ascending aorta at the proximal insertion site and coursing superiorly to the anastomosis with the native artery.

temporary arterial access, such as for cardiopulmonary bypass during aortic arch repair or placement of a temporary ventricular support device (37). Alternative access is necessary when the ascending aorta cannot be used because of severe atherosclerosis, dissection, or aneurysm. Subclavian artery access through side grafts is becoming increasingly common, because this method has a lower risk of arterial stenosis and injury compared with direct cannulation. When arterial access is no longer required, the anastomosis between the graft and artery is left intact and the end of the graft is oversewn or tied off, creating the appearance of a stump (36).

Postsurgical changes from subclavian artery bypass and side grafts can simulate pseudoaneurysms or contrast material extravasation that may be seen in vascular injury. Bypass grafts often have a thickened irregular contour that may be mistaken for vasculitis or vascular injury. At noncontrast CT, graft material is typically hyperattenuating in comparison with normal vascular walls. Side branch graft remnants often have a squared-off or pointed distal portion, with sharp

Figure 30. Subclavian-subclavian arteriovenous graft in a 41-year-old man with end-stage renal disease who required hemodialysis and had a history of multiple failed arteriovenous grafts in the upper extremities. (A) Axial MPR CT image shows an arteriovenous graft in the left chest wall, connecting the subclavian vein and ipsilateral subclavian artery (arrow). (B) Three-dimensional reconstruction CT image shows the graft arising from the subclavian vein, looping anteriorly, and crossing superiorly to the ipsilateral subclavian artery anastomosis (arrowhead).



margins and hyperattenuating suture material or pledgets at the distal aspect to reinforce the closure (37) (Fig 31). Acquiring noncontrast and/or delayed portal venous phase images in addition to arterial phase images is essential to distinguish these expected postsurgical changes from true subclavian abnormalities.

Knowledge of the patient's clinical presentation and surgical history and evaluation of the surrounding tissues for evidence of injury or prior surgical procedure are helpful when assessing whether a finding is a normal postprocedural finding or a pathologic entity. Postprocedural changes should be considered in the presence of adjacent surgical clips, suture material, or median sternotomy wires. Recognizing altered anatomy also helps to guide the radiologist to look for additional evidence of an arterial bypass or ligation.

Conclusion

The subclavian artery is an important structure that may be involved in a variety of abnormalities, may serve as a collateral vessel, or may have a role in a range of vascular procedures. Familiarity with normal and variant anatomy and expected postprocedural findings provides a foundation for determining if the imaging appearance is within normal limits and helps the radiologist to distinguish a true subclavian artery abnormality from potential mimics. Knowledge of subclavian artery abnormalities also helps to narrow the differential diagnosis and ultimately directly affects patient care and outcomes.

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Figure 31. Postsurgical changes of prior right subclavian artery cannulation in a 46-year-old man with a history of type A aortic dissection and repair of the ascending aorta. Axial CT image acquired after the surgery shows a tubular protuberance (arrow), with a squared-off edge extending from the anterior margin of the subclavian artery that represents a short segment of the residual side-graft material from the surgery.

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